

AIMS Customer GUI Design Specifications

Project: AIMS (An Internet Media Store) Software **Module:** Customer Interface **Scope:** Functional Requirements, Screen Content, and Transitions (Visual styles excluded)

1. Customer Screens (IDs + Purpose)

These are the primary boundary classes (screens) identified for the customer user actor.

- **S1 – Product List:** The entry point when opening AIMS. It displays 20 random products initially and supports searching, filtering by category/price, and paging.
- **S2 – Product Details:** Displays the full specification of a single product. The specific fields displayed vary dynamically based on the media type (Book, Newspaper, CD, or DVD).
- **S3 – Cart:** Displays items currently selected for purchase, quantities, prices (excluding VAT), and stock availability warnings. Allows the user to update quantities or remove items.
- **S4 – Delivery Information:** A form where the user enters their personal and shipping details. The system validates required fields and calculates the specific delivery fee based on total weight and location.
- **S5 – Payment / Invoice:** Displays the final invoice (Product total + VAT + Shipping) and handles the payment process via default QR code or optional Credit Card input.
- **S6 – Order Confirmation:** The final success screen showing the order summary and transaction details after a successful payment. The order status is set to "Pending Processing."

Note: Generic error pop-ups (validation errors, stock errors, payment timeouts) are treated as "dependent child screens" (Pattern 2).

2. Screen Content (Wireframe-Level Description)

S1 – Product List

Main Goal: Browse, search, filter, sort products, and add items to the shopping cart.

- **Search & Filter Section:**
 - **Text Box:** "Search by title"
 - **Dropdown:** "Category" selection
 - **Price Range Filter:** Options such as <100k, 100–200k, 200–300k, etc.
 - **Button:** "Search"
- **Product List Area:**

- *Displayed as a grid or list of repeating cards:*
- **Thumbnail Image**
- **Title**
- **Category**
- **Short Description**
- **Current Price** (displayed excluding VAT)
- **Action:** "Add to cart" button (with quantity input field)
- **Action:** "View details" link/button
- **Pagination:**
 - Page numbers and Next/Previous controls.
- **Global Actions:**
 - **Button/Icon:** "View Cart" (Navigation to S3)

S2 – Product Details

Main Goal: View full product information and add specific quantities to the cart.

- **Product Summary:**
 - Title, Category, Price, Stock Status.
 - General Description, Physical Dimensions, Weight.
- **Type-Specific Details (Dynamic Area):**
 - **Book:** Authors, Cover Type, Publisher, Publication Date, Number of Pages, Language, Genre.
 - **Newspaper:** Editor-in-Chief, Publisher, Publication Date, Issue Number, Frequency, ISSN, Sections.
 - **CD:** Artists, Record Label, Track List (Titles + Lengths), Genre, Release Date.
 - **DVD:** Disc Type, Director, Runtime, Studio, Language, Subtitles, Release Date, Genre.
- **Actions:**
 - **Input:** Quantity Selector.
 - **Button:** "Add to cart".
 - **Button:** "Back to list" (Navigation to S1).
 - **Button:** "View Cart" (Navigation to S3).

S3 – Cart

Main Goal: Review selected items, manage quantities, handle stock availability issues, and proceed to checkout.

- **Cart Summary List:**
 - **Product Name** (Clickable link returning to S2).
 - **Unit Price** (excl. VAT).
 - **Quantity** (Editable input).
 - **Line Subtotal.**
 - **Action:** "Remove item" button.

- **Total Product Price** (excl. VAT).
- **Stock Warning Section:**
 - *Conditional Display:* If an item request exceeds available stock, show message: "Only [X] items available for [ProductName]".
- **Actions:**
 - **Button:** "Update cart" (Recalculates totals and re-checks stock).
 - **Button:** "Continue shopping" (Navigation back to S1).
 - **Button:** "Place order" (Navigation to S4).

S4 – Delivery Information

Main Goal: Collect delivery details, validate inputs, and calculate shipping fees.

- **Cart Recap (Read-Only):**
 - List of items, quantities, and current subtotal (excl. VAT) for reference.
- **Delivery Form:**
 - **Customer Name** (Required).
 - **Phone Number** (Required).
 - **Address** (Street, District).
 - **Province/City** (Dropdown - Logic distinguishes Hanoi/HCM vs. others for fees).
 - **Delivery Method** (Standard/Express options, if applicable).
- **Fee Calculation Display:**
 - Total weight of items.
 - Calculated Shipping Fee.
 - *Info Note:* Explanation of free shipping rules (e.g., "Free up to 25,000 VND if order value > 100,000 VND").
- **Validation:**
 - Inline error messages for missing or invalid fields.
- **Actions:**
 - **Button:** "Recalculate shipping" (Can be automatic on field change or manual).
 - **Button:** "Back to cart" (Navigation to S3).
 - **Button:** "Continue to payment" (Navigation to S5).

S5 – Payment / Invoice

Main Goal: Display final invoice and process payment via QR or Credit Card.

- **Invoice Details:**
 - List of items, quantities, line totals.
 - Total product price (excl. VAT).
 - **VAT** (10%).
 - Total product price (incl. VAT).
 - **Shipping Fee.**
 - **Grand Total** (Final amount to be paid).
- **Payment Method Section:**

- *Selection Mechanism (Tabs or Radio Buttons):*
- **Option A: QR Code (Default)** - Displays dynamic QR image + instructions.
- **Option B: Credit Card** - Form for Card Number, Expiry, CVV, Name.
- **Adjustment Options:**
 - Link: "Change delivery info" (Return to S4).
 - Link: "Change items" (Return to S3; triggers invoice recalculation).
- **Actions:**
 - **Button:** "Pay now" (for QR) or "Pay with card".
 - *Error State:* Timeout or failure messages with "Retry" or "Change method" options.

S6 – Order Confirmation

Main Goal: Confirm successful transaction and display order details.

- **Order Information:**
 - Customer Name, Phone, Shipping Address, Province.
 - Total Amount Paid.
- **Transaction Details:**
 - Transaction ID.
 - Transaction Content/Description.
 - Transaction Datetime.
- **Order Status:**
 - Display status as: "Pending Processing."
 - *Info Note:* "An email containing the invoice and payment info has been sent."
- **Actions:**
 - **Button:** "Back to store / Continue shopping" (Navigation to S1).

3. Screen Transition Notations

These notations define the navigational flow and logic triggers between the screens defined above.

- **S1 (Search/Filter/Sort/Paging) → S1**
 - *User remains on Product List while refining view.*
- **S1 (Select Product) → S2**
 - *User clicks a specific item to view details.*
- **S2 (Back to List) → S1**
- **S1 or S2 (View Cart) → S3**
- **S3 (Update Qty / Remove Item) → S3**
 - *Cart refreshes to show new totals.*
- **S3 (Continue Shopping) → S1**
- **S3 (Place Order, Stock OK) → S4**
 - *Proceeds only if requested quantities ≤ available stock.*

- **S3 (Place Order, Stock Not OK)** $\rightarrow$ **S3 + Stock Warning Popup**
- **S4 (Confirm, Valid Info)** $\rightarrow$ **S5**
- **S4 (Confirm, Invalid Info)** $\rightarrow$ **S4 + Validation Popup**
- **S4 (Back to Cart)** $\rightarrow$ **S3**
- **S5 (Switch Payment Method)** $\rightarrow$ **S5**
 - *Toggles between QR view and Credit Card form.*
- **S5 (Payment Success)** $\rightarrow$ **S6**
- **S5 (Payment Failure / Timeout)** $\rightarrow$ **S5 + Error Popup**
- **S5 (Change Delivery)** $\rightarrow$ **S4**
- **S5 (Change Items)** $\rightarrow$ **S3**
- **S6 (Back to Store)** $\rightarrow$ **S1**